electronics (for circuit boards). When exact (large) solutions require too long to be computed, simulated annealing can be a viable alternative, providing near-optimal results in a reasonable time.

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[1] Technically even gradient descent can move past a local minimum, depending on the shape of the function around it and on the learning rate: see figure 16.15.

[2] The original name would be *hill climbing*, when the goal of the optimization is maximization; since we aim to reduce cost, though, that would be confusing.

[3] Not having a static formula can be helpful with dynamic cost functions, for instance, externally-computed metrics, or simulations (for instance, in reinforcement learning the cost is determined by running a simulation.

[4] Greedy algorithms make locally-optimal choices (for instance, they only go to positions with lower cost); unfortunately this doesn't always translate into getting to optimal results.

[5] We can consider it a category of heuristics, aka *meta-heuristic*: each algorithm using simulated annealing to solve a specific problem is going to be a heuristic.

[6] With the right configuration and, as we'll see, with a small-range transition function.

[7] You can take a look at some JavaScript code as part of JsGraphs library, on GitHub.

[8] Accordingly, the constant k is called Boltzmann's constant.

[9] We described hill descent in chapter 16.

[10] We would have to constrain the first and last delivery in each city to be the to be the closest to the connections to the previous and next city in the tour. Even so, though, it's possible that the solution find in 2 steps is not the best possible: if the connections to other cities starts in different areas of town, this could influence the choice of the best sequence for the cities; the influence, however, could be small enough to be considered acceptable.

[11] As we have discussed in chapter 16, NP-hardness is based on the worst-case scenarios: many problems, however, are difficult only for a minority of edge cases, while many real-world scenarios can be tackled more efficiently.

[12] Or, equivalently, we could say that we were dealing with precisely one embedding on the graph…

[13] Because we are restricted to using straight-line segments, that are a sub-set of all possible curves for edges.

[14] P. Eades. A heuristic for graph drawing. Congressus Numerantium, 42:149-160, 1984.

[15] T. Kamada and S. Kawai. An algorithm for drawing general undirected graphs. Information
Processing Letters, 31, 1989.